

Template for Computer Programming I Practical Recording

Date	:	16/01/2025
Problem Specification	:	Calculate the area and perimeter of the circle.
Assumption	:	The user input is valid integer.
Limitation	:	N/A
Input	:	Input the radius of the circle
Processing	:	The area of the circle is calculated using the formula: $\text{area} = \pi * r * r$, AND The perimeter of the circle is calculated using the formula: $\text{perimeter} = 2 * \pi * r$
Output	:	Display the area and perimeter of the circle
Algorithm	:	Step1: Prompt as "Enter the radius of the circle " Step2: Initialize variable r for radius Step3: Read the input r given by user Step4: Assign input value for r Step5: initialize area as $\text{area} \leftarrow (\pi) * r * r$ Step6: initialize perimeter as $\text{perimeter} \leftarrow 2 * \pi * r$ Step6: print value of area as "The area of the circle is: " Step7: Print value of perimeter as "The perimeter of the circle is: ". Step8: End the program.
Programme listing	:	program attached
Test data and expected output	:	1. Test data: $r=7$ Expected output: $\text{area} = 153.93804002589985$ $\text{Perimeter} = 43.982297150257104$
Output obtained for test data	:	1. Test data: $r=7$ Obtained output: $\text{area} = 153.93804002589985$ $\text{Perimeter} = 43.982297150257104$
Discussion	:	Discuss the challenges or take away note for this program

Conclusion

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Summarize the problem it aims to solve.